DATA SCIENCE & STATISTICS NOTES

Z-Score & Probability Fundamentals

Complete technical summary and practical implementation reference for GitHub log repository.

Source: Sheryians AI School — *Complete Statistics Course for Beginners* (Instructor: Akarsh Vyas)
Tracked Timestamps: 01:38:09 – 01:54:44 (Z-Score & Density Curves) | 01:54:44 – 02:01:15 (Basic Probability)

1. Standard Score (Z-Score) Analysis

A Z-score measures how many standard deviations an individual data point lies away from the population or sample mean. It normalizes distinct distributions onto a standardized scale with a mean of 0 and a standard deviation of 1.

> **MATHEMATICAL FORMULATION**
>
> $$Z = (X - \mu) / \sigma \quad | \quad Z = (X - \bar{x}) / s$$
>
> Where **X** is the raw observation, **μ** (or $\bar{x}$) is the mean, and **σ** (or **s**) is the standard deviation.

Key Applications of Z-Scores

- **Outlier Detection:** Data points with a absolute Z-score threshold $|Z| > 3.0$ (or 2.5 depending on context) are typically flagged as statistical anomalies.
- **Feature Standardization:** Machine learning algorithms reliant on distance metrics (e.g., KNN, SVM, K-Means, Gradient Descent optimization) require feature scaling via Z-score transformation.
- **Cross-Distribution Comparison:** Allows direct comparison of data points from entirely different measurements (e.g., SAT vs. ACT scores).

2. Fundamentals of Probability Theory

Probability quantifies the likelihood of an event occurring, bounded strictly within the numerical range of **$0 \le P(A) \le 1$**.

Concept	Formula / Rule	Core Definition
Classical Probability	$P(A) = n(A) / n(S)$	Ratio of favorable outcomes to total sample space elements.
Complement Rule	$P(A') = 1 - P(A)$	Probability that event A does NOT occur.
Addition Rule (Or)	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	Probability that event A or event B (or both) occur.
Multiplication Rule (And)	$P(A \cap B) = P(A) \times P(B\|A)$	Joint probability of both events occurring sequentially or together.
Conditional Probability	$P(A\|B) = P(A \cap B) / P(B)$	Likelihood of event A given that event B has already occurred.

3. Probability Distributions & Empirical Rule

In a standard normal distribution (Gaussian Curve), data points align according to the **Empirical 68–95–99.7 Rule**:

- **68.27%** of values lie within **1 Standard Deviation** (μ ± 1σ)
- **95.45%** of values lie within **2 Standard Deviations** (μ ± 2σ)
- **99.73%** of values lie within **3 Standard Deviations** (μ ± 3σ)

4. Python Implementation Snippet

```
      import numpy as np import scipy.stats as stats # Sample Dataset data = np.array([12, 15, 14, 10, 10, 100, 11, 13, 14]) # Note outlier: 100 # Calculate Z-Scores mean = np.mean(data) std = np.std(data) z_scores = [(x - mean) / std for x in data] # Outlier Detection Threshold outliers = [data[i] for i in range(len(z_scores)) if abs(z_scores[i]) > 2.0] print(f"Detected Outliers: {outliers}")
```